# Simple Algebra

Year 4 · **Lesson Worksheet**

## Worked Examples

**A) Easy:** Find the missing number: 7 + □ = 12.

**Answer:** 12 − 7 = 5. Answer: 5.

**B) Standard:** A function machine adds 3, then doubles the result. If the input is 4, what is the output?

**Answer:** (4+3) = 7, then 7×2 = 14.

**C) Advanced:** Find the missing number: 2 × □ + 5 = 17.

**Answer:** 2 × □ = 12, so □ = 6.

## Easy / Standard (15 questions)

1. Find the missing number: 5 + □ = 11.
2. Find the missing number: □ − 4 = 9.
3. Find the missing number: 3 × □ = 18. A) 5 B) 6 C) 7 D) 8
4. Find the missing number: □ ÷ 5 = 4.
5. Find the missing number: 20 − □ = 8.
6. Select ALL the number sentences below where the missing number is 6: A) □ + 5 = 11 B) □ + 4 = 12 C) 2 × □ = 12 D) □ − 3 = 9 E) 18 ÷ □ = 3
7. Find the missing number: □ + 9 = 15.
8. Find the missing number: 6 × □ = 30. A) 4 B) 5 C) 6 D) 7
9. Find the missing number: □ ÷ 3 = 7.
10. Find the missing number: 14 + □ = 20.
11. Find the missing number: □ − 7 = 5.
12. Select ALL the number sentences below where the missing number is 4: A) □ × 3 = 12 B) □ + 6 = 12 C) 20 ÷ □ = 5 D) □ − 1 = 5 E) 9 + □ = 13
13. Find the missing number: 4 × □ = 28.
14. A function machine adds 5, then doubles. If the output is 16, what was the input? A) 3 B) 6 C) 8 D) 11
15. Find the missing number: □ + 15 = 22.

## Standard (15 questions)

1. A function machine adds 2, then multiplies by 3. If the input is 4, what is the output?
2. A function machine subtracts 3, then doubles. If the input is 10, what is the output?
3. A function machine multiplies by 2, then adds 5. If the input is 6, what is the output? A) 15 B) 16 C) 17 D) 18

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4. Find the missing number: 2 × □ + 3 = 15.
5. Find the missing number: □ ÷ 2 + 4 = 9.
6. Select ALL of these where the missing number is 5: A) 3 × □ = 15 B) □ + 8 = 12 C) 20 − □ = 15 D) □ × 4 = 16 E) 2 × □ + 1 = 9
7. Find the missing number: 3 × □ − 4 = 11.
8. A function machine adds 7, then halves. If the input is 9, what is the output?
9. Find the missing number: 4 × □ + 2 = 26. A) 5 B) 6 C) 7 D) 8
10. Find the missing number: □ ÷ 4 − 1 = 3.
11. A function machine doubles, then adds 6. If the output is 20, what was the input?
12. Find the missing number: 5 × □ − 3 = 17.
13. Select ALL of these where the missing number is 7: A) □ + 5 = 12 B) 3 × □ = 24 C) 2 × □ − 4 = 10 D) □ − 2 = 5 E) 21 ÷ □ = 3
14. Find the missing number: 6 × □ + 5 = 41.
15. A function machine adds 4, then triples. If the output is 30, what was the input? A) 5 B) 6 C) 7 D) 8

## Standard / Hard (15 questions)

1. Find the missing number: 2 × (□ + 3) = 16.
2. Find the missing number: (□ − 3) × 2 = 14. A) 7 B) 8 C) 9 D) 10
3. I think of a number, add 4, then double it. The answer is 20. What was the number?
4. Find the missing number: 3 × □ − 5 = 16.
5. Select ALL of these where the missing number is 6: A) 2 × □ − 2 = 10 B) □ + 10 = 16 C) 4 × □ = 20 D) 30 − □ = 24 E) 3 × (□ − 1) = 15
6. A rectangle has a perimeter of 20cm. If the length is 6cm, what is the width? (perimeter = 2 × (length + width))
7. Find the missing number: 3 × □ + 2 × □ = 20.
8. I think of a number, multiply it by 2, then add 3. The answer is 15. What is the number? A) 5 B) 6 C) 7 D) 8
9. Two numbers add to 16. One number is 2 more than the other. What are the two numbers?
10. Find the missing number: 4 × □ + 2 × □ = 24.
11. A box of pens costs £2 more than a box of pencils. One box of pens and one box of pencils cost £12 in total. How much does a box of pencils cost?
12. Select ALL of these where the missing number is 5: A) 2 × (□ + 1) = 12 B) 3 × □ − 2 = 13 C) 2 × □ + 5 = 15 D) 4 × □ = 20 E) (□ + 3) × 2 = 18
13. Find the missing number: 6 × □ − 2 × □ = 20.
14. A number is doubled, then 5 is added. The result is 19. What is the number? A) 6 B) 7 C) 8 D) 9
15. A function machine adds 6, then subtracts 2. If the input is 5, what is the output?

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